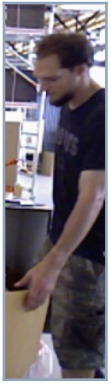
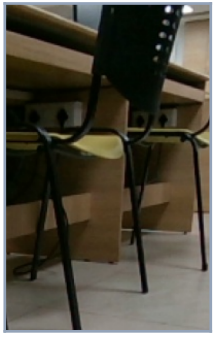
# Qualitative Mobility & Physical Affordance Case Studies (Actual Observation Crops)



### (a) Self-Mobile Agent (Person)

Object: Person
GT Affordance: MOBILE (Self-Mobile)
GT Motion: MOVING
VLM: MOVABLE (Conf: 0.80)
World Disp: 1.42 m
Cam Apparent: 0.88 m

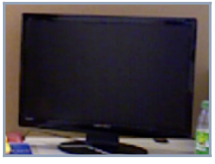
ACTION: Mask dynamic distractor



### (b) Relocated Object (Chair)

Object: Chair
GT Affordance: MOVABLE
GT Motion: MOVING
VLM: MOVABLE (Conf: 0.80)
World Disp: 0.52 m
Cam Apparent: 0.61 m

ACTION: Invalidate historical anchor



### (c) Stationary Item (Monitor)

Object: Monitor
GT Affordance: MOVABLE
GT Motion: STATIONARY
VLM: IMMOVABLE (Conf: 0.90)
World Disp: 0.03 m (Stationary)
Cam Apparent: 0.34 m

ACTION: RETAIN as stable landmark



### (d) Immovable Structure (Desk)

Object: Desk
GT Affordance: STRUCTURALLY_IMMOVABLE
GT Motion: STATIONARY
VLM: IMMOVABLE (Conf: 0.95)
World Disp: 0.01 m (Rigid)
Cam Apparent: 0.22 m

ACTION: Prioritize as mapping anchor